

INFORMATION AND COMMUNICATION THEORY

UEC-310

By
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Course Content

- **Module-1 : Probability Theory**

- Random Variable, Distribution function, Characteristic function and central limit theorem

- **Module-2 : Stochastic Processes**

- Systems with Stochastic Inputs, Markov Chains, and Random-Walk Model

- **Module-3 : Estimation & Hypothesis Testing**

- Hypothesis, Neyman Pearson Tests, Bayes' Criteria, Z-Score, and p-Value Test

- **Module-42 : Information Theory**

- Information Measure, Entropy, Source coding, Mutual Information, channel capacity, Shannon's Theorems.

- **Module-5 : Statistical Modeling of Noise**

- Wide-Sense-Stationary (WSS) Processes, Poisson Process Noise, Noise Statistics, Noise Power Spectral Densities,

Syllabus

UEC310:INFORMATION AND COMMUNICATION THEORY

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Course Objectives:

To gain knowledge and understand the concepts of probability theory, random variables, stochastic processes and Information theory. To familiarize the students with the applications of probabilistic/stochastic techniques/methods in communication engineering and information theory.

Details of Contents:

Probability Theory:

Review of Probability, Bayes' Theorem, Bernoulli Trials, Bernoulli's Theorem, Concepts of Random Variables, Distribution and Probability Density Functions, Binomial Random variables, Conditional Distributions, Functions of One Random Variable its Distribution, Mean and Variance, Moments, Characteristic Functions, Two Functions of Two Random Variables, Joint Moments, Joint Characteristic Functions, Conditional Distributions, Conditional Expected Values, Normality, Central Limit Theorem.

Stochastic Processes:

Systems with Stochastic Inputs, Power Spectral Analysis of I/O Signals, Poisson Points, Cyclostationary Processes, Poisson Sum Formula, Ergodicity, Mean Square Estimation, Markov Chains, and Random-Walk Model

Estimation & Hypothesis Testing:

Time and Ensemble Averages, Covariance Functions. Simple Binary Hypothesis Tests, Decision Criteria, Neyman Pearson Tests, Bayes' Criteria, z-Score, and p-Value Test

Information Theory:

Introduction, Information measure and entropy, Information source, Markov source, Adjoint of an information source, Joint and Conditional Information measure, Joint and conditional information measure of a Markov source, Instantaneous codes, Kraft-McMillan inequality, Shannon first theorem, Coding strategies and Huffman coding, Introduction to information channels, mutual information and channel capacity, Shannon second theorem, channel capacity calculations for different channels, Differential entropy, Rate distortion theory.

Statistical Modeling of Noise:

Probability Density of a Jointly-Gaussian Random Vector, Wide-Sense-Stationary (WSS) Processes, Poisson Process Noise, Noise Statistics in Linear Time-Invariant Systems, Noise Power Spectral Densities, Signal-to-Noise-Ratio in Presence of AWGN and Interferences.

Text Books:

1. Athanasios Papoulis, Probability Random Variables and Stochastic Processes, McGraw-Hill (1984)
2. John N. Daigle, Queueing Theory with Applications to Packet Telecommunication, Springer (2005)
3. Bernard Sklar, Digital Communications: Fundamentals and Applications, Prentice Hall (2001)

Reference Books:

1. P.Z. Peebles, Probability, Random Variables, and Random Signal Principles, McGraw-Hill (1980)
2. Dimitri P. Bertsekas, Robert G. Gallager, Data Networks, Prentice-Hall (1987)
3. A. Larson and B.O. Schubert, Stochastic Processes, vol. I and II, Holden-Day (1979)
4. W. Gardener, Stochastic Processes, McGraw Hill (1986)
5. IEEE Transactions on Information Theory
6. David J. C. Mackay, "Information Theory, Inference and Learning Algorithms", Cambridge University Press, 2003

PROBABILITY THEORY

Lecture – 1	Introduction to Probability Theory
Lecture – 2	Bernoulli Trials and Bernoulli's Theorem
Lecture – 3	Random Variables
Lecture – 4	Conditional Distribution and Bayes' theorem
Lecture – 5	Function of a Random Variable
Lecture – 6	Mean and Variance of random variable
Lecture – 7	Moments and Characteristic Function
Lecture – 8	Function of Two Random Variables
Lecture – 9	Joint Moments and Joint Characteristic Functions
Lecture – 10	Conditional Distributions, Conditional Expected Values
Lecture – 11	Central Limit Theorem

INTRODUCTION

Experiment: to describe any process that generates a set of data.

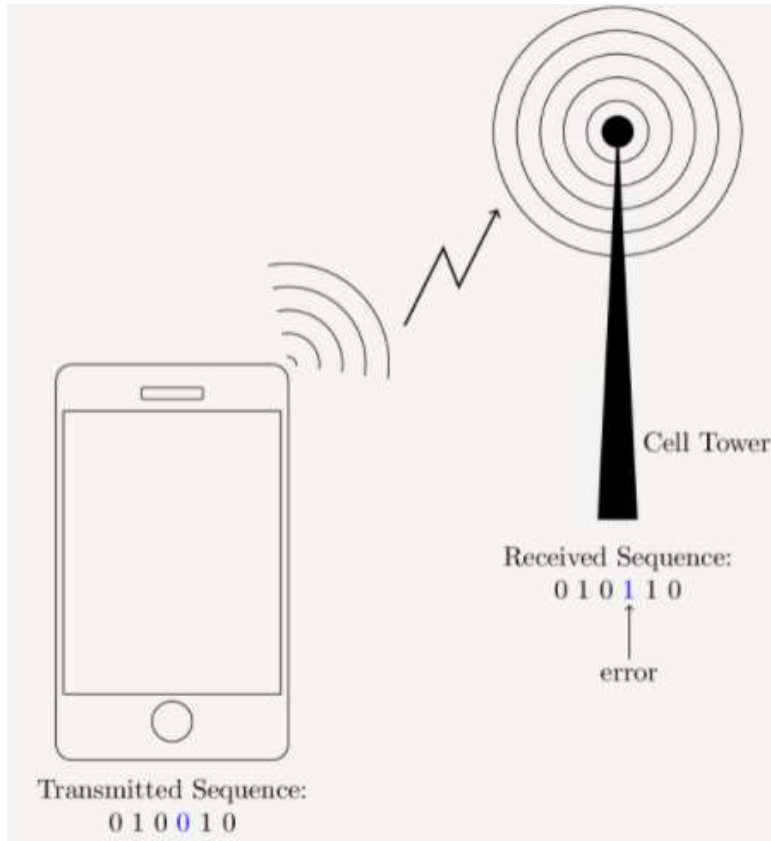
- ***Tossing of a coin.*** In this experiment, there are only two possible outcomes, heads or tails.
- ***Rolling a dice.*** In this experiment, there are six possible outcomes, surface count 1 to 6.
- ***Launching of a missile*** and observing of its velocity at specified times.

Interest in the **observations** obtained by **repeating the experiment** several times.

The **outcomes** will depend on **chance** and, therefore, cannot be predicted with **certainty**.

RANDOMNESS

Examples



Mobile communication

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Flipping a coin



Rolling a dice

Probability theory deals with the study of random phenomena, which under repeated experiments yield different outcomes.

The set of all possible outcomes of a statistical experiment is called the **sample space** and is represented by the symbol Ω .

Each outcome in a sample space is called an **element** or a **sample point**.

EXAMPLE: The **sample space** of possible outcomes when **two coins are tossed**, may be written as

$$\Omega = \{ \xi_1, \xi_2, \xi_3, \xi_4 \}.$$

And elements/sample points are

$$\xi_1 = (H, H), \quad \xi_2 = (H, T), \quad \xi_3 = (T, H), \quad \xi_4 = (T, T)$$

where H and T correspond to heads and tails, respectively.

When an **experiment** is performed under these conditions, certain elementary events ξ_i occur in different but ***completely uncertain ways***.

We can assign nonnegative number $P(\xi_i)$, as the probability of the event ξ_i .

Laplace's Classical Definition: The Probability of an event A is defined a-priori without actual experimentation as

$$P(A) = \frac{\text{Number of outcomes favorable to } A}{\text{Total number of possible outcomes}},$$

provided all these outcomes are ***equally likely***.

Example-1.1:

Consider a box with ***n* white** and ***m* red** balls. In this case, there are two elementary outcomes: white ball or red ball. Probability of “***selecting a white ball***”

$$= \frac{n}{n + m}.$$

Example-1.2:

In a rolling dice experiment, there are six elementary outcomes: ***n=3*** for even and ***m=3*** for odd numbers. The probability of “***getting a even number***” :

$$= \frac{n}{n + m} = \frac{3}{3 + 3} = \frac{1}{2}.$$

Even Number: (e.g. 2, 4, 6).

The totality of all outcomes $\xi_i, (xi)$ *known a priori*, constitutes a **whole set** Ω , the set of all experimental **outcomes**.

ξ_i = ith Outcome

$$\Omega = \{ \xi_1, \xi_2, \dots, \xi_k, \dots \}$$

Ω has **subsets** $A, B, C, \dots$. Recall that if A is a **subset** of Ω , then if $\xi \in A$ implies $\xi \in \Omega$. From A and B , we can generate other related subsets $A \cup B, A \cap B, \bar{A}, \bar{B}$, etc.

$$A \cup B = \{ \xi \mid \xi \in A \text{ or } \xi \in B \}$$

$$A \cap B = \{ \xi \mid \xi \in A \text{ and } \xi \in B \}$$

and

$$\bar{A} = \{ \xi \mid \xi \notin A \}$$

Assuming that the probability $p_i = P(\xi_i)$ of elementary outcomes ξ_i of Ω are *apriori* defined, *how does one assign probabilities to more 'complicated' events such as $A \cup B$, $A \cap B$, $\bar{A}$, $\bar{B}$, etc., above?*

The **three axioms of probability** (defined on the next slide) can be used to achieve that goal.

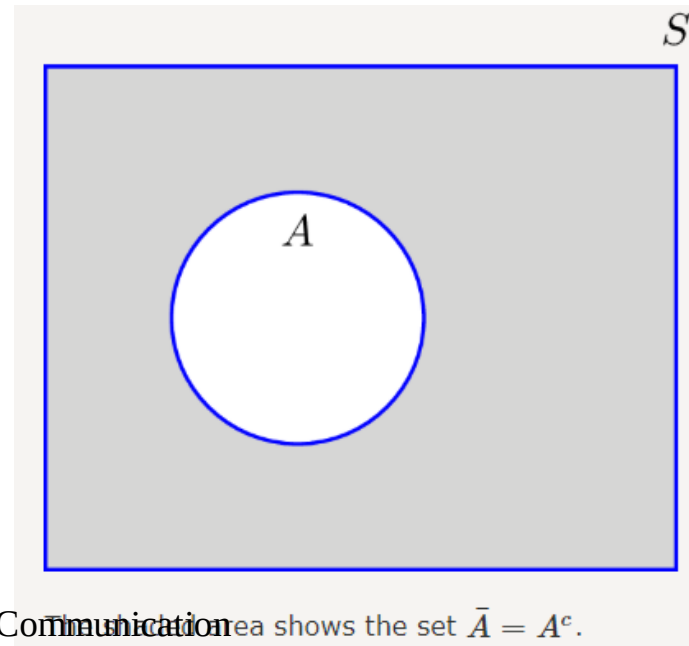
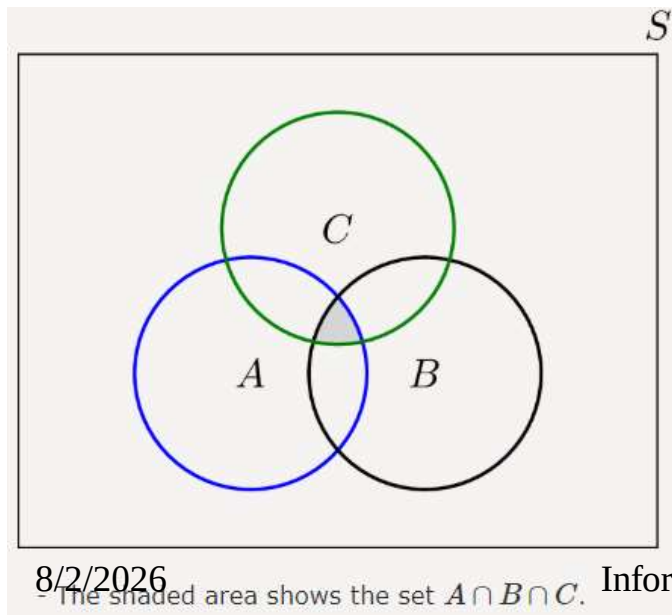
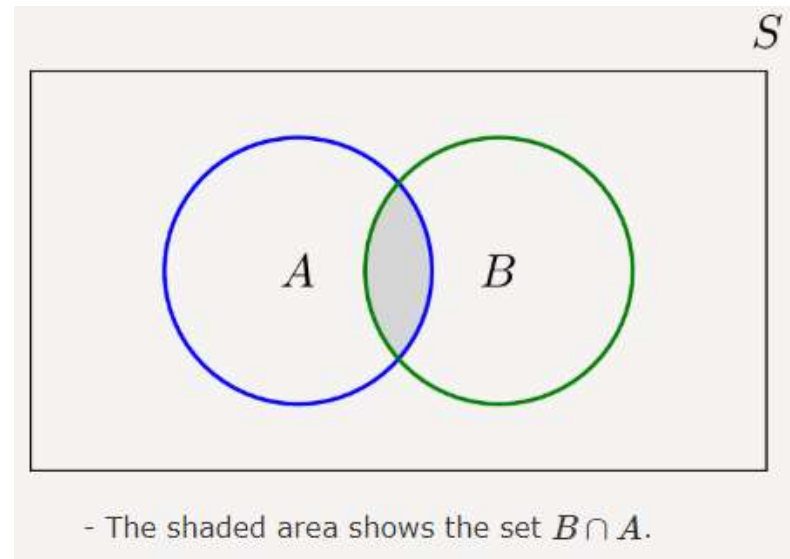
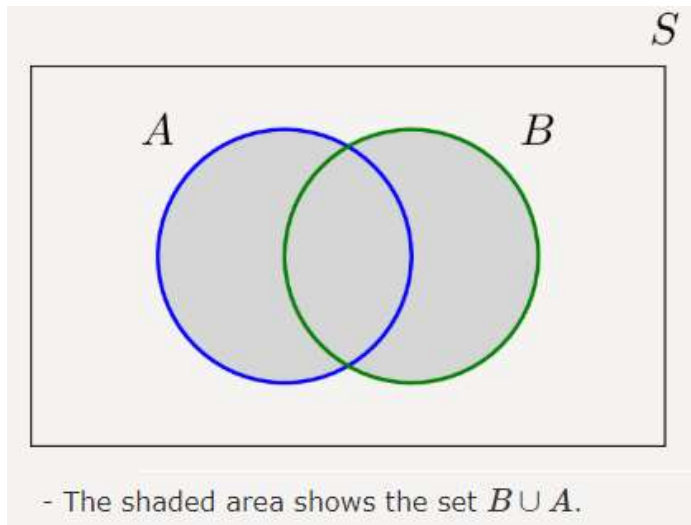
Axioms of Probability

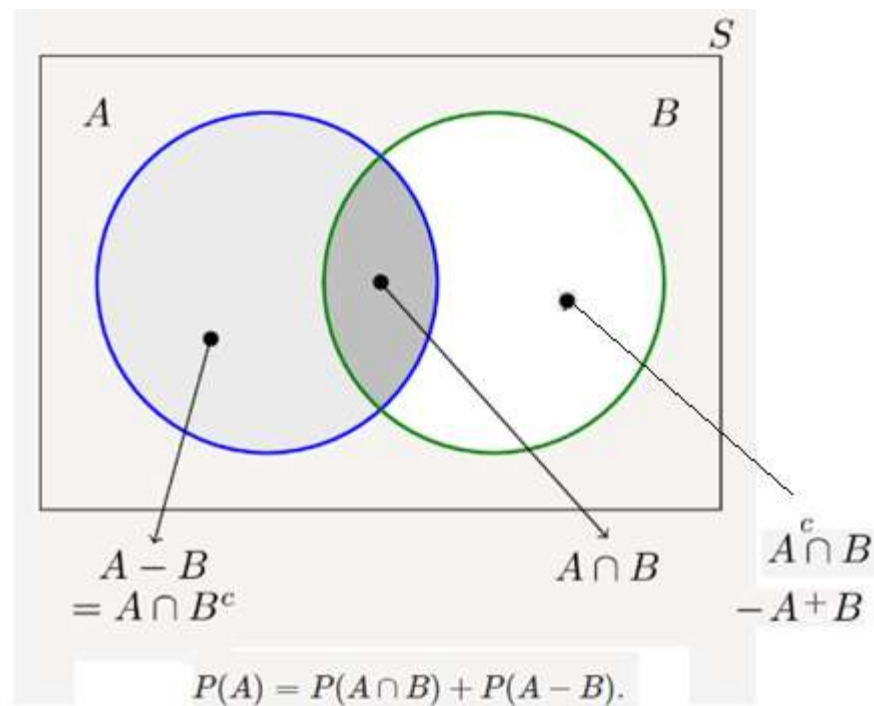
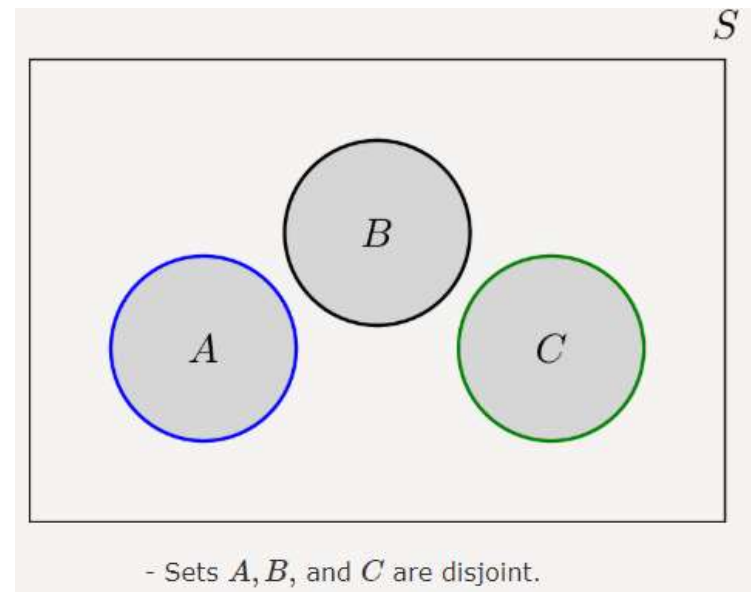
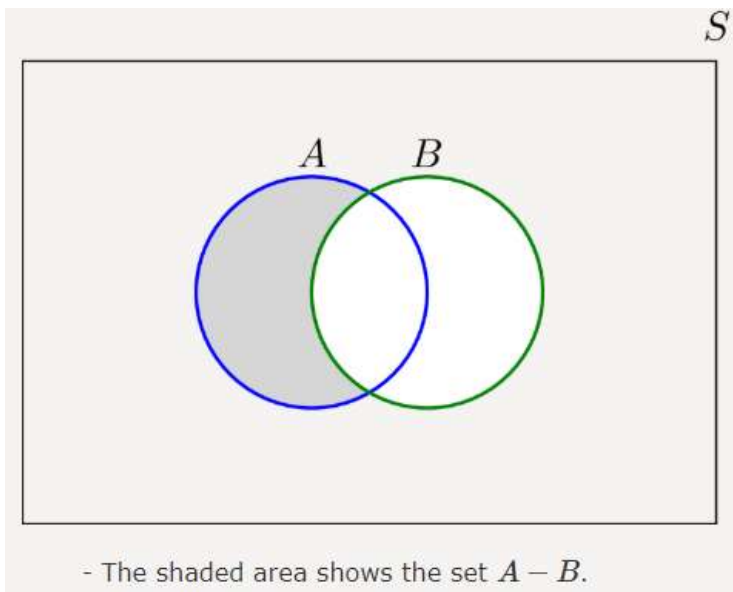
For any **event** A , a number $P(A)$ is assigned, called the **probability** of the event A . This number satisfies the following three conditions that act as the **axioms of probability**.

- (i) $P(A) \geq 0$ (Probability is a nonnegative number)
- (ii) $P(\Omega) = 1$ (Probability of the whole set is unity)
- (iii) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$.

*(Note that (iii) states that if A and B are **mutually exclusive** (M.E.) events, the probability of their union is the sum of their probabilities.)*

Probability Theory: Set Theory





Probability Theory

Axioms of Probability:

- Axiom 1: For any event A , $P(A) \geq 0$.
- Axiom 2: Probability of the sample space S is $P(S) = 1$.
- Axiom 3: If $A_1, A_2, A_3, \dots$ are disjoint events, then $P(A_1 \cup A_2 \cup A_3 \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$

Notation:

- $P(A \cap B) = P(A \text{ and } B) = P(A, B),$
- $P(A \cup B) = P(A \text{ or } B).$

Theorem 1.1: De Morgan's law

For any sets $A_1, A_2, \dots, A_n$, we have

- $(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n)^c = A_1^c \cap A_2^c \cap A_3^c \dots \cap A_n^c$;
- $(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n)^c = A_1^c \cup A_2^c \cup A_3^c \dots \cup A_n^c$.

Theorem 1.2: Distributive law

For any sets A, B , and C we have

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Inclusion-Exclusion Principle:

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$,
- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

- Axiom 1: For any event A , $P(A) \geq 0$.
- Axiom 2: Probability of the sample space S is $P(S) = 1$.
- Axiom 3: If $A_1, A_2, A_3, \dots$ are disjoint events, then $P(A_1 \cup A_2 \cup A_3 \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$

a. **Since** $A \cup \bar{A} = \Omega$, we have using (ii)

$$P(A \cup \bar{A}) = P(\Omega) = 1.$$

But $A \cap \bar{A} \in \phi$, and using (iii),

$$P(A \cup \bar{A}) = P(A) + P(\bar{A}) = 1 \quad \text{or} \quad P(\bar{A}) = 1 - P(A).$$

b. **Similarly**, for any A , $A \cap \{\phi\} = \{\phi\}$.

Hence it follows that $P(A \cup \{\phi\}) = P(A) + P(\phi)$.

But $A \cup \{\phi\} = A$, and thus $P\{\phi\} = 0$.

c. **Suppose** A and B are *not mutually exclusive* (M.E.)?

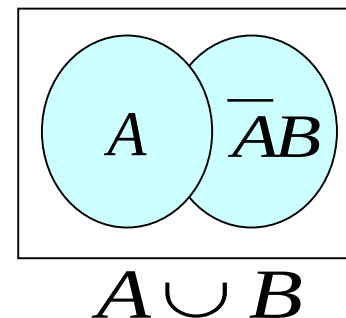
How does one compute $P(A \cup B) = ?$

To compute the above probability, we should re-express $A \cup B$ in terms of M.E. sets so that we can make use of the probability axioms.

$$A \cup B = A \cup \bar{A}B,$$

where A and $\bar{A}B$ are clearly M.E. events.

Thus using axiom



$$P(A \cup B) = P(A \cup \bar{A}B) = P(A) + P(\bar{A}B). \quad (1.1)$$

To compute $P(\bar{A}B)$, we can express B as

$$B = B \cap \Omega = B \cap (A \cup \bar{A})$$

$$\text{Thus} \quad = (B \cap A) \cup (B \cap \bar{A}) = BA \cup B\bar{A}$$

$$P(B) = P(BA) + P(B\bar{A}), \quad (1.2)$$

since $BA = AB$ and $B\bar{A} = \bar{A}B$ are M.E. events.

From (1.2),

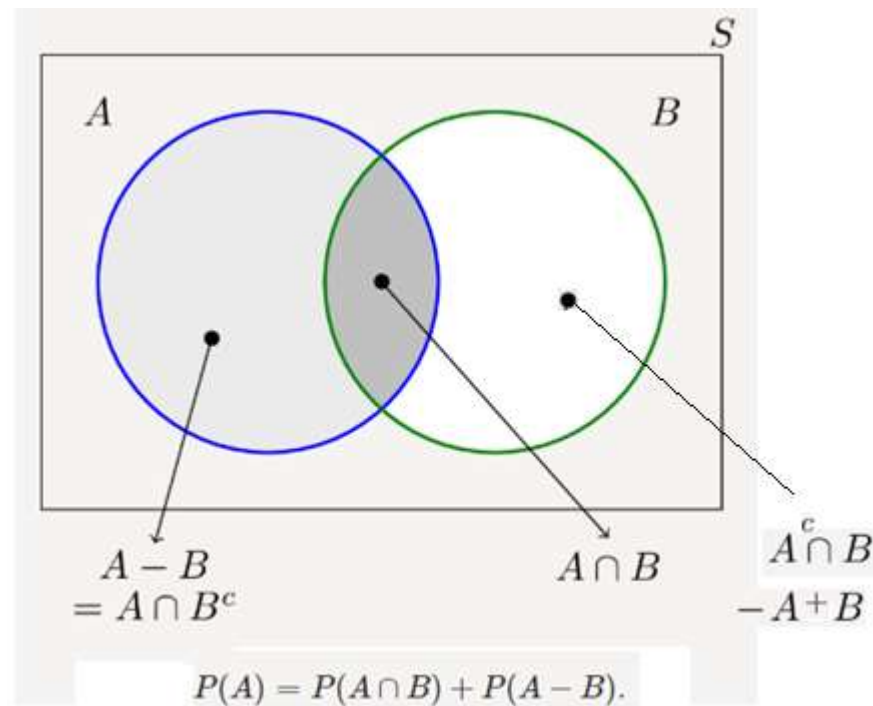
$$P(B) = P(BA) + P(B\bar{A}),$$

$$P(\bar{A}B) = P(B) - P(AB) \quad (1.3)$$

and using (1.3) in (1.1)

$$P(A \cup B) = P(A) + P(B) - P(AB).$$

Example:



Conditional Probability

If you obtain **additional information**, how should you **update** probabilities of events?

For example, suppose that in a certain city, **23 %** of the days are **rainy**. Thus, if you pick a random day, the **probability** that it rains that day is **23 %**: $P(R)=0.23$, where **R** is the **event** that it rains on the randomly chosen day.

Now suppose that I pick a **random day**, but I also tell you that it is **cloudy** on the **chosen** day. Now that you have this **extra piece** of information, **how** do you **update the chance** that it rains on that day?

In other words, what is the **probability** that it **rains** given that it is **cloudy**?

If **C** is the event that it is **cloudy**, then we write this as **$P(R|C)$** , the **conditional** probability of **R** given that **C** has occurred.

It is reasonable to assume that in this example, **$P(R|C)$** should be **larger** than the **original** $P(R)$, which is called the **prior probability** of **R**.

But **what** exactly should **$P(R|C)$** be?

Example:

I **roll** a fair die. Let A be the event that the outcome is an **odd number**, i.e., $A=\{1,3,5\}$.

Also let B be the event that the outcome is **less than or equal** to 3, i.e., $B=\{1,2,3\}$.

What is the probability of A , $P(A)$? What is the probability of A given B , $P(A|B)$?

Solution:

This is a finite sample space, so

$$P(A) = \frac{|A|}{|S|} = \frac{|\{1, 3, 5\}|}{6} = \frac{1}{2}.$$

Now, let's find the conditional probability of **A given that B occurred**.

If we know **B has occurred**, the outcome must be among **{1,2,3}**.

For **A to also happen** the outcome must be in **$A \cap B = \{1,3\}$** .

Since all die rolls are **equally likely**, we argue that **$P(A|B)$** must be equal to

$$P(A|B) = \frac{|A \cap B|}{|B|} = \frac{2}{3}.$$

Now let's see how we can generalize the above example. We can rewrite the calculation by dividing the numerator and denominator by $|S|$ in the following way

$$P(A|B) = \frac{|A \cap B|}{|B|} = \frac{\frac{|A \cap B|}{|S|}}{\frac{|B|}{|S|}} = \frac{P(A \cap B)}{P(B)}.$$

Although the above calculation has been done for a finite sample space with equally likely outcomes,

If A and B are two events in a sample space S , then the **conditional probability of A given B** is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ when } P(B) > 0.$$

Conditional Probability and Independence

In N independent trials, suppose N_A , N_B , N_{AB} denote the number of times events A , B and AB occur respectively.

$$P(A) \approx \frac{N_A}{N}, \quad P(B) \approx \frac{N_B}{N}, \quad P(AB) \approx \frac{N_{AB}}{N}.$$

Among these, N_A occurrences of A , and N_B occurrences of B .

only N_{AB} of them are also found among the N_B occurrences of B .

$$P(A|B) = \frac{N_{AB}}{N_B} = \frac{N_{AB} / N}{N_B / N} = \frac{P(AB)}{P(B)}$$

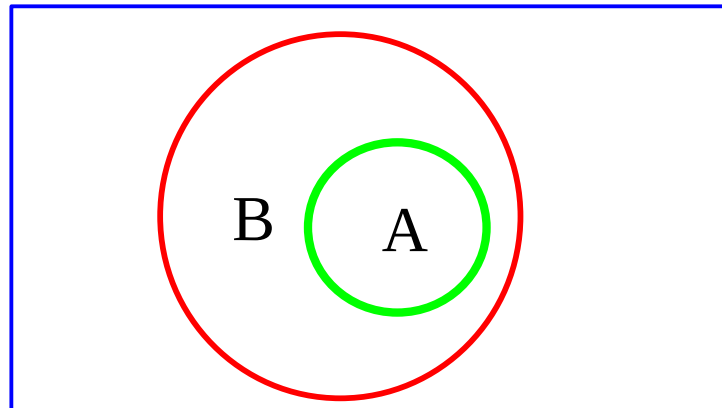
$P(A|B)$ = Probability of “the event A given that B has occurred”

$$P(A|B) = \frac{P(AB)}{P(B)},$$

Help: Basics

We say that set A is a proper subset of set B when every element of A is also an element of B, but there is at least one element of B that is not in A.

$A \subset B$
‘A is a proper subset of B’



Properties of Conditional Probability:

If $B \subset A$, $AB = B$, and

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

since if $B \subset A$, then occurrence of B implies automatic occurrence of the event A .

Example: Dice tossing experiment

$A = \{\text{outcome is even}\}$, $B = \{\text{outcome is 2}\}$,

a. If $B \subset A$, and $P(A | B) = 1$.

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

b. If $A \subset B$, $AB = A$, and

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{P(A)}{P(B)} > P(A).$$

Independence: A and B are said to be **independent** events, if

$$P(AB) = P(A) \cdot P(B).$$

$$P(BA) = P(B) \cdot P(A).$$

$$P(AB) = P(AB)$$

Notice that the above definition is a **probabilistic** statement, *not* a set **theoretic notion** such as **mutually exclusiveness**.

Suppose A and B are independent, then

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A).$$

Thus if A and B are **independent**, the event that B has occurred **does not add** any more **light** into the event A .

Example: *A box contains 6 white and 4 black balls. Remove two balls at random without replacement. What is the probability that the first one is white **and** the second one is black?*

Solution:

Let W_1 = “first ball removed is white”

B_2 = “second ball removed is black”

Continued...

We know that

Using the conditional probability rule,

$$P(W_1 B_2) = P(W_1 | B_2) P(B_2).$$

$$P(A | B) = \frac{P(AB)}{P(B)},$$

$$P(B_2 W_1) = P(B_2 | W_1) P(W_1).$$

For W_1 & B_2 to be independent

$$P(W_1 B_2) = P(B_2 W_1)$$

$$P(W_1) P(B_2) = P(B_2 W_1)$$

Verify the independency?.

Lets start with $P(B_2 W_1)$:

$$P(W_1) = \frac{6}{6+4} = \frac{6}{10} = \frac{3}{5},$$

and

$$P(B_2 | W_1) = \frac{4}{5+4} = \frac{4}{9},$$

and hence

$$P(B_2 W_1) = \frac{3}{5} \cdot \frac{4}{9} = \frac{12}{45} \approx 0.267$$

Are the events W_1 and B_2 independent?

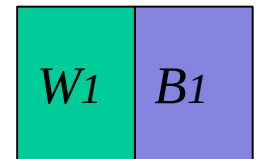
Our common sense says **No**.

To **verify** this we need to compute $P(B_2)$.

Of course the fate of the second ball very much depends on that of the first ball.

The first ball has two options: W_1 = “first ball is white” or B_1 = “first ball is black”.

Note that $W_1 \cap B_1 = \phi$, and $W_1 \cup B_1 = \Omega$.



Hence W_1 together with B_1 form a partition.

Let $W = \text{White ball} = 6$; $B = \text{Black ball} = 4$

Thus

$$\begin{aligned} P(B_2) &= P(B_2 | W_1)P(W_1) + P(B_2 | B_1)P(B_1) \\ &= \frac{4}{5+4} \cdot \frac{6}{4+6} + \frac{3}{6+3} \cdot \frac{4}{6+4} \\ &= \frac{4}{9} \cdot \frac{3}{5} + \frac{1}{3} \cdot \frac{2}{5} = \frac{4+2}{15} = \frac{2}{5}, \end{aligned}$$

Now , Test the independency-

$$P(B_2)P(W_1) = \frac{2}{5} \cdot \frac{3}{5} \neq P(B_2W_1) = \frac{12}{45}$$

As expected, the events W_1 and B_2 are dependent.

Concept	Meaning	Formulas
Disjoint	A and B cannot occur at the same time	$A \cap B = \emptyset,$ $P(A \cup B) = P(A) + P(B)$
Independent	A does not give any information about B	$P(A B) = P(A), P(B A) = P(B)$ $P(A \cap B) = P(A)P(B)$

Bayes' Theorem. From, $P(A|B) = \frac{P(AB)}{P(B)},$

$$P(AB) = P(A|B)P(B).$$

Similarly,

$$P(B|A) = \frac{P(BA)}{P(A)} = \frac{P(AB)}{P(A)},$$

or

$$P(AB) = P(B|A)P(A).$$

$$P(A|B)P(B) = P(B|A)P(A).$$

or

$$P(A|B) = \frac{P(B|A)}{P(B)} \cdot P(A)$$

(1.4)

Equation (1.4) is known as **Bayes' theorem**.

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

Bayes' theorem has an interesting interpretation:

$P(A)$ represents the *a-priori probability* of the event A . Suppose B has occurred, and assume that A and B are **not independent**.

How can this new information be used to update our knowledge about A ?

Bayes' rule take into account the new information (“ B has occurred”) and gives out the *a-posteriori probability* of A given B .

A more general version of **Bayes' theorem** is -

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{P(B)} = \frac{P(B | A_i)P(A_i)}{\sum_{i=1}^n P(B | A_i)P(A_i)},$$

Example:

Two boxes B_1 and B_2 contain 100 and 200 light bulbs respectively. The first box (B_1) has 15 defective bulbs and the second 5. **Suppose** a box is selected at **random** and one bulb is picked out.

(a) **What** is the probability that it is **defective**?

(b) **Suppose** we **test** the bulb and it is found to be defective. **What** is the probability that it came from **box-1**?

Solution:

Note that box B_1 has **85** good and **15** defective bulbs. Similarly box B_2 has **195** good and **5** defective bulbs.

Let D = “Defective bulb is picked out”.

Then

$$P(D | B_1) = \frac{15}{100} = 0.15, \quad P(D | B_2) = \frac{5}{200} = 0.025.$$

Since a box is selected at random, let us assume, they are equally likely.

$$P(B_1) = P(B_2) = \frac{1}{2}.$$

Thus B_1 and B_2 form a partition

$$\begin{aligned} P(D) &= P(D | B_1)P(B_1) + P(D | B_2)P(B_2) \\ &= 0.15 \times \frac{1}{2} + 0.025 \times \frac{1}{2} = 0.0875. \end{aligned}$$

Thus, there is **8.75%** probability that a bulb picked at random is **defective**.

*(b) Suppose we test the bulb and it is found to be defective.
What is the probability that it came from box-1?*

$$P(B_1 | D) = ?$$

$$P(B_1 | D) = \frac{P(D | B_1)P(B_1)}{P(D)} = \frac{0.15 \times 1/2}{0.0875} = 0.8571. \quad (1.5)$$

Notice that initially $P(B_1) = 0.5$; when we picked out a box at random and tested a bulb that turned out to be defective.

*Can this information shed some light about the fact that we might have picked up **box 1**?*

Observation:

From (1.5), $P(B_1 | D) = 0.857 > 0.5$, and indeed it is more likely at this point that we must have chosen box 1 in favor of box 2.

Similarly, $P(B_2 | D) = 0.1482 < P(B_1 | D)$, **WHY ?**

(**Recall** box1 has **six** times more **defective** bulbs compared to box2).

Reference

- Athanasios Papoulis, “Probability, Random Variables, and Stochastic Processes,” 3rd edition, McGraw Hill Publication.